Do Now · retrieval

Programming Techniques · Iteration and Tracing

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What prints after int x=2; x=x*3;?

Q2

What does a queue remove first?

Q3

Why should loops have a stopping condition?

Q4

What is 0 XOR 1?

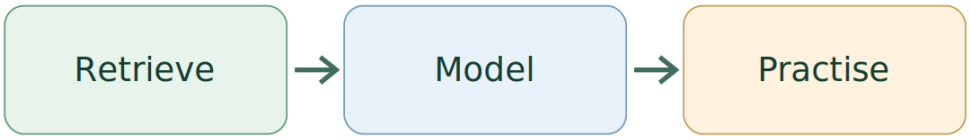
New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Explain the important decision — Initialise the accumulator once. Each iteration changes total; i++ then advances the counter. Test i <= 3 before every pass.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

iteration — Repetition of statements

accumulator — Running total

counter — Tracks repetitions

condition — Controls loop continuation

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Trace an accumulator

Follow the worked method

1. Initialise and trace pass one — Initially total=0 and i=1. The test 1<=3 is true; total becomes 1, then i becomes 2.

2. Trace the remaining passes — At i=2, total becomes 1+2=3; i becomes 3. At i=3, total becomes 3+3=6; i becomes 4.

3. Show the failed test — 4<=3 is false, so the body stops and total remains 6. A while loop may run zero times; a do...while tests after its body and runs at least once.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Discuss, then respond
for i = 0 to 3 exclusive, how many iterations?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Apply independently

Trace the final total for 2 + 4 + 6 using an accumulator. [3]

Question 2

Apply independently

Give a reason to choose a while loop rather than for. [2]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Apply independently

Write a C# loop to output 1, 2, 3. [3]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

True or false: a while loop always runs at least once.

Use key vocabulary from memory

1. Explain iteration in your own words.

2. Explain accumulator in your own words.

3. Explain counter in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.